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TOTAL ERROR IN A BIG DATA WORLD: ADAPTING THE TSE FRAMEWORK TO BIG DATA

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While Big Data offers a potentially less expensive, less burdensome, and more timely alternative to survey data for producing a variety of statistics, it is not without error. The AAPOR Task Force on Big Data and others have called for researchers to evaluate the quality of Big Data using an approach similar to the total survey error (TSE) framework. However, differences in the construction of, access to, and overall data structure between survey data and Big Data make application of TSE difficult. In this article, we seek to develop the Total Error Framework (TEF), an extension of the TSE framework, to be (1) more inclusive and applicable to many types of Big Data, (2) comprehensive in that it considers "total" error, and (3) unified in that it allows researchers to compare errors in Big Data to errors in survey data. After outlining this framework, we then illustrate an application of TEF by comparing error in housing unit area (square footage) estimates collected in a survey (the 2015 Residential Energy Consumption Survey [RECS]) to those estimates found in three Big Data databases (Zillow.com, Acxiom, and CoreLogic).

KEYWORDS: Administrative data; Big Data; Total error framework; Total survey error; RECS.

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1. INTRODUCTION

The use of Big Data to supplement or replace survey data is growing. In a world of escalating survey costs, falling response rates, and rising potential for survey biases, Big Data often offer a less expensive, less burdensome, and more timely alternative for producing a variety of statistics from health prevalence to consumer confidence to household characteristics (O'Connor, Balasubramanvan, Routledge, and Smith 2010; Butler 2013; Citro 2014). However, Big Data are not without their weaknesses; they suffer from their own sources of error, access challenges, and confidentiality concerns (Japec, Kreuter, Berg, Biemer, Decker, et al. 2015).

The AAPOR Task Force on Big Data and others have called for researchers to evaluate the quality of Big Data using an approach similar to the total survey error (TSE) framework (Groves and Lyberg 2010; Japec et al. 2015; Schober, Pasek, Guggenheim, Lampe, and Conrad 2016; Hsieh and Murphy 2017). The TSE framework enumerates each type of survey error and suggests that each should be evaluated individually and in combination to assess the net error on survey estimates (Groves 2004; Biemer 2010; Groves and Lyberg 2010). By understanding the primary types of error, researchers may adjust the survey design, processing steps, and analytic methods to minimize or compensate for it. When supplementing survey data with Big Data, understanding the primary types of error found within each data source allows researchers to capitalize on the strengths from each and create a stronger, combined product (Biemer 2016a). Understanding the strengths and weaknesses of both data sources would also help identify what circumstances and research questions may be best answered by each data source.

Several researchers have investigated error in Big Data, but they have frequently focused on one-off evaluations instead of outlining a framework that can be applied to many types of Big Data. For example, Ladouceur, Rahme, Pineau, and Joseph (2007) investigated how varying assumptions on the quality of administrative health records biased estimates of osteoarthritis prevalence among Quebec citizens. Similarly, data from social media sites such as Facebook and Twitter have been analyzed for their ability (or, in some cases, inability) to predict election outcomes (e.g., Ceron, Curini, Iacus, and Porro 2014; Kalampokis, Karamanou, Tambouris, and Tarabanis 2017; Kennedy, Wojcik, and Lazer 2017). Finally, several articles have been written to explain why the Google Flu Trends estimates were both unbiased when first created and biased once algorithms and search habits changed (Ginsberg, Mohebbi, Patel, Brammer, Smolinski, et al. 2008; Cook, Conrad, Fowlkes, and Mohebbi 2011; Butler 2013).

Some researchers have decomposed the error from Big Data sources into its components, but this research has neither been comprehensive in its coverage of error types nor inclusive of the range of Big Data sources. In outlining the weaknesses of Big Data, Baker (2017) points out that Big Data were not originally collected for the researcher's intended purpose and may suffer from undercoverage (coverage error) and item-missingness (nonresponse/missing error). He also notes that Big Data analytics are frequently automated. Without human intervention, critical thinking and quality control, data scientists may place too much confidence in the resulting models, leading them to draw incorrect inferences (analytic error). Schober et al. (2016) opted to limit their discussion to error found in social media analyses. Specifically, they cite three primary differences between survey and social media: (1) participants do not understand the activity in which they are engaged in the same way, (2) the nature of the data differs, and (3) ethical differences in the data generation step (e.g., informed consent versus webscraping). The authors map out several examples of the differences, including how the sources of error may also differ. For example, measurement/content error (as defined later in our article) may be larger when analyzing social media content (as opposed to survey data) because individuals have no expectation of privacy and want to portray their best self. While these are valid and important comments from the authors, they do not systematically examine each error component or its application to Big Data.

Hsieh and Murphy (2017) have attempted to create an error framework that may be used to assess the quality of Twitter data. They identified three types of error in Big Data: (1) coverage error, (2) query error, and (3) interpretation error. Under their definitions, coverage error includes both overcoverage and undercoverage of both users and tweets. Overcoverage refers to the inclusion of units in the analytic data set that are not in the target or inferential population and should be excluded while undercoverage refers to the exclusion of units in the analytic data set that are in the target or inferential population and should be included. Query error, which is a type of coverage error, occurs when the researchers use keywords to scrape data from the Twittersphere. Using too many generic words may result in the inclusion of tweets irrelevant to the research question, whereas using incomplete search terms may lead to the failure to capture a complete picture. Finally, interpretation error occurs when coding tweets in a manner as to create variables that may be used in analysis. While the Hsieh and Murphy framework is a good start at decomposing error sources in Big Data, the framework is specific to one type of Big Data (Twitter); it excludes some types of error (e.g., data processing and analytic errors), conflates others (e.g., query error may conflate specification error, coverage error, and measurement/content error), and does not easily allow for direct comparisons between survey and Big Data since their framework does not directly map to the TSE framework.

Finally, Biemer (2016b) restructured the traditional TSE model in which error components are enumerated on the type of error and refocused the discussion on where the error may be found in a standard rectangular data set: (1) row, (2) column, or (3) cell. For example, missing units (e.g., household, person, transaction, or trip) would typically be ascribed to coverage error, sampling error, or nonresponse error in the TSE framework, depending on the reason for missingness. Since each row is typically a unique unit, missing units would attribute to row error under Biemer's framework. The simplicity of this approach makes it applicable to any type of structured data source. However, it is not directly applicable to unstructured data that do not conform to a row/column/cell format. Although these unstructured data may ultimately be converted to a rectangular format, that conversion process would also be subject to errors, and those errors are not explicitly represented in Biemer's framework. One benefit of representing these preprocessing errors is that it helps to illuminate the root causes of errors, which is important for purposes of error mitigation and control.

In this article, we seek to overcome the previous limitations by outlining how the TSE framework may be extended to comprehensively identify and classify errors found in Big Data. The resulting adaptation should be (1) more inclusive and applicable to many types of Big Data, (2) comprehensive in that it considers "total" error, and (3) unified in that it allows researchers to compare errors in Big Data to errors in survey data. After outlining this framework, we then illustrate an application of the framework by comparing error in housing unit area (square footage) estimates collected in a survey (the 2015 Residential Energy Consumption Survey (RECS)) with those estimates found in three Big Data databases (Zillow.com, Acxiom, and CoreLogic).

2. A TOTAL ERROR FRAMEWORK FOR BIG DATA

This section adapts the TSE framework to Big Data. This more general framework can be used to identify, describe, and interpret the errors in a structured or unstructured Big Data set—be it massive or small, static or dynamic. We refer to this framework as the "Total Error Framework" (TEF), cutting the "S" ("survey") from "TSE." A variety of researchers have provided their own set of error components for the TSE framework and its predecessors. For example, Deming (1944) (though not referencing the TSE framework by name) outlined thirteen factors associated with errors, and Groves, Fowler, Couper, Singer, and Tourangeau (2004) more recently enumerated six components. Similarly, individuals may conceptualize the research process in different ways. For the purposes of this article, we enumerate eight error components that make up the TEF and consider the collection and data analysis processes in chronological order (figure 1).

The left set of boxes outlines the steps unique to the process when using survey data; the right set of boxes includes the steps unique to the process when using Big Data; and the center set of boxes includes steps found regardless of the data source. We assume the chronology followed for survey data is familiar to the reader and do not go into detail here.

Although Big Data analysis is often exploratory, we assume without much loss of generality that a specific research question is to be addressed in the analysis. For Big Data, the steps are (1) identifying the appropriate data

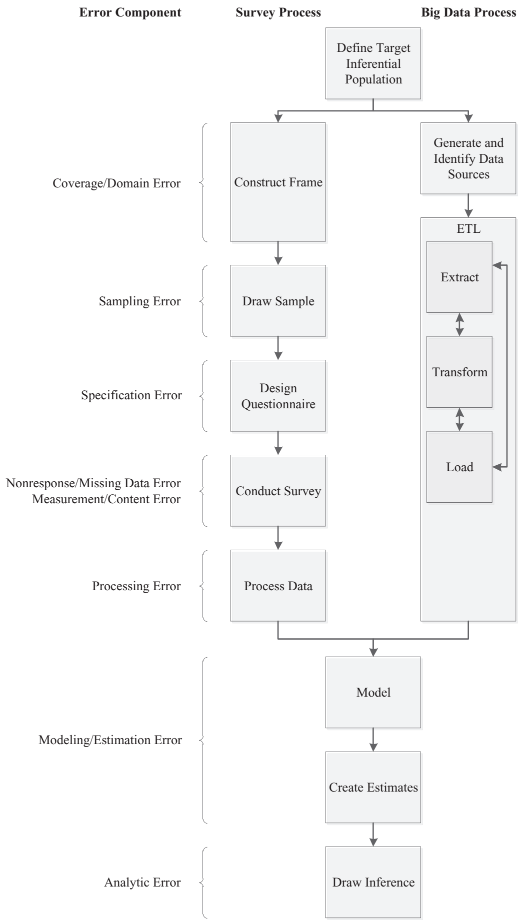

Figure 1. Data Process.

source(s) for the analysis and (2) extracting, transforming, and loading the data into a database appropriate for further analysis. These two steps are somewhat unique to creating a Big Data set. Data source identification is similar to survey frame identification and construction. However, instead of identifying the appropriate lists to acquire and potentially merge for survey sampling, the researcher considers what Big Data may already exist that could be used to provide insights to the research question. Examples of preexisting data sources include Twitter data or "tweets," which may be used to answer questions about how individuals interact with each other or to identify influencers; administrative records such as electronic health records, which may be used to assess treatment protocols of various conditions or monitor vaccination rates; transactional databases such as IT problem logs to identify system weaknesses and improve workforce efficiencies; or video images from traffic cameras that could be used to assess traffic patterns, violations, or unsafe road conditions. Countless other examples of data sources range from light pollution maps to Google search histories to data collected through health monitors (e.g., FitBit). Additional examples of preexisting Big Data sources are provided throughout the text and still more can be found in a variety of books on the subject (e.g., Salganik 2017; Stephens-Davidowitz 2018).

Step two includes several substeps: extraction, transformation, and loading. Researchers infrequently use all of the data within a given source for a variety of reasons. They may have a specific domain (e.g., time period) in which they are interested. For example, researchers using Twitter data to analyze the 2016 US presidential election would not want to review all tweets since its inception but would instead launch a query to identify and extract tweets made between January and November 2016 that included researcher-specified key terms (e.g., Trump or Clinton or election). This is similar to the way in which researchers would subset the United States national address-based frame to relevant ZIP Codes when conducting a state survey. Alternatively, researchers may not have the computing power or data storage requirements, or they may be limited by licensing or cost agreements, requiring them to draw a sample. For example, the data use agreement for the Zillow.com API, which we discuss at some length in section 3, limits downloads to 1,000 addresses per day, which often requires the researcher to implement sampling. Regardless of the reason, researchers may extract a portion of the Big Data for use in their analyses.

The extracted data may then be transformed to create a data set that is appropriate for analysis. Transformation can take many forms and will be dependent on the original data structure. Let us assume a simple case such as transaction data in which every item sold in a store makes up its own row in the Big Data source. Assuming the researcher desires to conduct item-level analysis, no transformation is necessary. For a more complex example, assume the data are traffic camera images. The images would need to be transformed into a data set by coding the images for the required information (e.g., speed or number of cars on the road). Similarly, tweets about an election may need to be coded as either favorable or unfavorable of each candidate. The tweets may also need to be reformatted to a different unit of analysis—for example, individual-level data. If the goal is to estimate the proportion of persons that have favorable tweets about a candidate but the data are stored by tweet, person-level data must be created by some transformation process. Frequently, unstructured or structured Big Data may require restructuring to conform to a traditional, rectangular file so that traditional data analysis can proceed. This is especially true when data for population units are being combined across multiple data sources.

Finally, the transformed data are loaded and stored on the researcher's specified server.

While it is helpful to define each of these substeps independently, in reality they often cannot be parsed. A singular system may be used to implement the

Table 1. Error Causes by Error Component and Data Source

TEF error component	Error causes	
	Survey data	Big Data
Coverage error (Unit or Within unit)	Undercoverage Overcoverage Duplication	Undercoverage Overcoverage Duplication
Sampling error	Inadequate sample sizes for analyses of subdomains or nonkey variables to detect significant differences Weight variation	Large probability samples of the full data set often lead to statistically significant differences
Specification error	Concepts and definitions for some constructs are difficult to operationalize	Concepts and definitions inconsistent with available data
Nonresponse/ missing data error	Item nonresponse Unit nonresponse	Item missingness Unit missingness
Measurement/ content error	Unintentional errors Deliberate errors	Unintentional errors Deliberate errors
Processing error	Editing errors Keying errors	Editing errors Incorrect data linkage
Modeling/ estimation error	Imputation Calibration/weighting Nonresponse adjustment	Imputation Calibration/weighting Missingness adjustment
Analytic error	Inferences to small areas and rare demographic groups often difficult Biases and measurement variances are often not considered in analyses	Often univariate and preclude most forms of quasi-experimental modeling requiring highly dimensional data Biases and measurement variances are often not considered in analyses

entire process effectively simultaneously, or researchers may use an iterative approach—using one method to extract the data, review it, then edit the process, and start again.

When using Big Data, the other steps (e.g., defining the inferential population) are similar to those taken when using survey data and are not discussed here. Regardless of the data source, each component of the process from data generation to the final analysis contains some risk of error. Table 1 summarizes several reasons for error by TEF error component and data source, but the specific reasons will vary by use case. The remainder of this section further describes each of these error components and causes. Because of the plethora of literature already published on the causes of error of survey data (for a thorough review, see Biemer 2010 or Groves et al. 2004) and because the procedures used to model, create estimates, and draw inference are similar for both survey and Big Data, we have focused the majority of our discussion on the causes of error that are unique to Big Data.

2.1 Coverage Error

For survey frames, coverage error can include undercoverage, overcoverage, and duplicative frame entries. Undercoverage refers to the failure of some target population units to be listed on the frame. This may result in coverage bias depending on the magnitude of the undercoverage and the difference between covered and uncovered population units for the characteristics of interest. Overcoverage occurs when out-of-scope units (e.g., vacant housing units in a survey of households) are found on the frame or when duplicates are included (e.g., multiple phone numbers linked to the same household or person). Overcoverage results in inefficiencies and may increase costs. Duplicates, if not identified, can lead to bias in that the duplicated units will be over-represented in the estimates.

Frame error may also occur within unit. A simple example is within dwelling unit coverage error where some individuals that live at a sampled address may be missed during the within-household rostering process. However, within-unit coverage errors are more general; examples include undercoverage of purchases in an expenditure survey, automobile trips in a transportation survey, television watching events in an audience media survey. Subsetting the population into domains can also lead to coverage error (for example, including only establishments in a few major industries on the frame for a survey of hazardous waste management practices).

All of the aforementioned coverage concerns may be the result of the source of the database from which the frame was derived (e.g., the United States Computerized Delivery Sequence file for address-based frames) or introduced by the researcher. For example, the landline random-digit dial (RDD) database includes all possible landline telephone numbers in the United States. However, the researcher may opt to subset the database to 1+ listed banks[1] prior to sample selection, ultimately introducing undercoverage by design.

Although there is no frame per se in Big Data analysis, the population represented by Big Data sources can be distorted in very similar ways by undercoverage, overcoverage, and duplication of the target population members at both unit and within-unit levels. For example, in an analysis of Twitter data,

1. RDD samples in the United States are typically drawn from a bank of 1,000 numbers. One draws a seven-digit number (a bank) from a list of active area codes and prefixes, then randomly generates the last three digits to create a phone number. When using 1+ banks, the list of area codes and prefixes from which a sample can be drawn is subset to combinations for which there is at least one listed telephone number. For more information on 1+ banks and coverage, see Boyle, Bucuvalas, Piekarski, and Weiss (2009).

the population represented by the data is Twitter users who are disproportionately younger adults. Thus, if inferences are to be made to the general population, older adults will be under-represented (i.e., undercovered). Likewise, overcoverage can occur to the extent that businesses maintain Twitter accounts (e.g., @RTI_Intl or @Walmart), whereas the focus of the research is on individuals. Businesses are not eligible and are thus overcovered. Duplication can arise when individuals have multiple Twitter accounts, and these cannot be identified in an analysis that focuses on individual Twitter users. When using multiple Big Data sources, the risk of duplicates may be higher because a common identifier may not be available. For example, one could use different email addresses or handles for different types of databases. Or identifiers on one frame (e.g., license plates of cars that traverse an intersection) may not be the same as another database (e.g., fuel purchases from gas stations).

Similar to survey data, the entirety of the Big Data set is rarely of interest to the researcher and must be subset to a specific domain. In many instances, the extraction process may have employed purposive or convenience subsetting, which may introduce coverage error. For example, pollsters attempting to predict the outcome of an election may extract data based upon relevant Google searches of key terms in a database of all searches. To the extent that such extractions may not include all relevant terms, may misspecify the time period (e.g., they do not include absentee votes), or may not include alternative spellings or typos, the resulting frame will not be representative. Likewise, the research may be interested in all tweets that a user sent regarding some topic during the specified time period. Within-unit undercoverage arises when only some of the tweets are identified due to query error (as defined by Hsieh and Murphy, 2017).

2.2 Sampling Error

We define sampling error to include errors that arise as a result of analyzing a (typically random) subset of the population of interest rather than the entire population (census). A number of factors can contribute to survey sampling error including the size of the sample, the sample design (e.g., stratification/clustering), and weight variation due to unequal selection probabilities. While most researchers set the target number of completed interviews based on the desired precision of key estimates, this does not ensure that precision will be adequate for all analyses of subdomains or other variables that were not used in the precision calculation. Inadequate sample size will increase the confidence intervals and reduce the precision of the resulting estimates. In addition, stratification and cluster sampling and unequal probability sampling will affect sampling error.

One of the advantages of Big Data is that they can be massive, and estimates derived from them may result in little or no sampling error. If a large probability sample of the data set is selected (e.g., to estimate model parameters), the large sample size nearly guarantees that a statistical test will detect a significant difference, even when no practical or meaningful difference exists on the full data set. Similar to large sample surveys, this situation may lead to false confidence in the accuracy of the sample-based estimates when large biases are present.

2.3 Specification Error

Specification error occurs when the concept (or construct) needed to address a research question does not precisely align with the concept implied by the data item. As an example, a key purpose of the Residential Energy Consumption Survey is to obtain the square footage of the housing unit. However, there is no universally accepted definition of what should be included in this calculation. Should it include only heated space (typically excluding the attic), closets, or unfinished space? The difference between the survey definition and the researcher's desired definition is attributed to specification error.

For many constructs of interest to social scientists, direct measures are not available and composite "proxy" measures are constructed from multiple survey items. For example, no single question can measure social engagement, so questions are instead asked about friends, activities, and memberships in which the respondent may engage. Likewise, it may be possible to combine a number of Big Data elements to measure social engagement (e.g., the number of followers on Twitter or friends on Facebook). However, the resulting measurement may not exactly agree with the researcher's concept of the construct. While data can be manipulated to create reliable indicators of some constructs of interest, the available variables in Big Data sets are typically predefined with little or no researcher control.

2.4 Nonresponse/Missing Data Error

For surveys, missing data is typically a consequence of nonresponse either at the unit or item level. Unit nonresponse occurs when a sample unit does not participate in the survey, and item nonresponse is the result of a respondent failing to provide an answer to a posed question. Missing data will lead to a bias that is the product of the missing data rate and the average difference in the feature between complete records and those with missing data (Lessler and Kalsbeek 1992).

Missing data in Big Data sets may be a consequence of a number of factors. For example, square footage for a housing unit may be missing in the Zillow.com database either because it has never been listed for sale, it was listed but the realtor failed to enter it, the database does not have access to a given county's record, or a value may have been entered incorrectly and deleted as implausible in the Big Data editing process. Thus, it would be a misnomer to classify all missing data in Big Data sets as nonresponse when that may be only one possible cause. It may be more appropriately labeled nonresponse/missing data. Regardless, missing data may introduce bias whether from nonresponse or other causes.

In Big Data sets, missing data are often confounded with undercoverage. What may qualify as nonresponse error in one analysis would qualify as coverage error in another. Let us take two examples of the same data source. Assume the research goal of a Twitter analysis is to identify changes in the perceptions of individuals in a population of the US president's performance over the course of his/her first 90 days in office. If the unit of analysis is the individual, then we may consider undercoverage to be the result of individuals who do not have Twitter accounts and nonresponse-related missingness to be Twitter subscribers who did not tweet about the president in his/her first 90 days in office. Given the same research question, assume the unit of analysis is the tweet, not the individual. Under this scenario, the appropriate frame unit is a tweet. All missingness could be ascribed to undercoverage, so both nonsubscribers and subscribers who chose not to tweet would be considered uncovered. Only if we knew the circumstances surrounding each missing tweet could we ascribe the missingness to either undercoverage or nonresponse.

Eckman and Kreuter (2017) argue this confounding effect is not unique to Big Data and can also be observed in survey data; they suggest assessing survey coverage bias and nonresponse bias together to avoid these complexities. This approach may be valid for evaluation, but to mitigate missing data, it is often better to understand the causes. We argue that the appropriate analysis plan for quantifying error in survey or Big Data is dependent on the data sources themselves. The important point is that each of these reasons for error are considered and categorized, the categorizations are transparent to others, and the resulting error is quantified and compared across data sources.

2.5 Measurement/Content Error

Measurement errors in surveys can arise from several causes that may accumulate to create substantial risks for bias. In addition, just as sampling error reduces the precision of the estimates, variable measurement errors can create noise in the data that decreases estimator precision. Measurement errors arise from ambiguous questions, imperfect memory of past events, interviewer influences, respondent comprehension errors, and even deliberate falsification by respondents, interviewers, and even organizations or institutions. For example, individuals may not report drug use because they do not want to be reported to the police, or they may underestimate their income because they forgot to include nonwage income.

In Big Data sets, erroneous data may be a consequence of a number of factors including the measurement process, transcription errors, data conversion errors, false readings from mechanical devices, outdated information, and so on. Content error is a more general term that captures all of these various causes of erroneous data in Big Data sets. For example, a real estate database may not include the new bedrooms that were just built as part of a new addition or the merging of two apartments into a single unit. Relatedly, a camera meant to count vehicles on the road may have difficulty determining whether a double-trailer semitruck is one vehicle or two. Both scenarios likely increase variance due to measurement error. Measurement error will bias estimates when the error is consistently in one direction. Bias may result when individuals tend to only present their best pictures on Instagram, portraying a much more attractive scene than may be reality. An inventory of bank transactions may underestimate illegal or promiscuous behavior if these types of activities are recorded using a less descriptive business name that cannot be categorized into an expense category.

2.6 Processing Error

Processing error for survey data includes errors arising during data entry, coding, editing, disclosure limitation, and variable conversions or transformations. Like measurement errors, they can contribute biases and increased variances in the estimation process.

Likewise, Big Data may suffer from many of the same errors during data processing. For example, data entry errors may arise during the data generation stage. Once the data are extracted, they may undergo a number of data processing steps toward producing a file for data analysis, including editing and transformation. The amount of information for any given record is much more limited for Big Data compared with survey data, potentially affecting the extent to which the researcher may make informed edits and increasing the risk of error. The extent of the required transformation depends on the difference between the original data layout and the layout needed for analysis. The more complex the transformation, the higher the risk of introducing processing error.

Another error that may arise, much more so than for survey data, occurs during the linking of Big Data records with population or survey sample members, which can be quite difficult and prone to error. Some units may have missing values on the variable used for linkage. Others may have outdated information (e.g., if an individual relocates, his/her address may have been updated on one data set but not the other), creating nonmatches or mismatches between files. Finally, the unit of the two data sets may be incompatible for linkage. While one data set includes a roster of individuals without addresses, another may only include addresses without names. A third source or probabilistic matching would need to be employed to link the two data sources.

2.7 Modeling/Estimation Error

Modeling error for both surveys and Big Data encompasses the errors arising from deficiencies in missing data and coverage error weighting adjustments, as well as imputation for item missing data. While similar types of models may be used for both survey and Big Data, they may have different risks of producing error. The missing data mechanism (i.e., the underlying stimulus that determines sample member participation) for Big Data is unknown and more difficult to model than for survey data. Thus, modeling and adjusting for missing data bias in the final estimates can be challenging. Similarly, and as described previously, the limited amount of information available for any given Big Data record also limits the variables available for imputing missing data. On the positive side, methods for compensating for nonresponse, missing data, and coverage errors in Big Data are possible using external calibration data (similar to methods applied to survey data). However, it is often challenging to accurately link data records across data sources at the person (or unit) level for the purposes of sample calibration or imputation.

2.8 Analytic Error

Analytic error includes errors made by data users and clients in analyzing and interpreting the results. As with processing and modeling procedures, similar methods may be applied to both survey and Big Data sets with differing risks for error. A strength of Big Data is that it has high spatial, temporal, and demographic granularity, which suggests that inferences to small areas and rare demographic groups usually inaccessible by survey inference are quite possible. For example, Zillow.com data include more than 110 million housing units that provide ample data for constructing precise estimates for small areas. In addition, Big Data provides opportunities to study the relationship between sample units (via social networks, for example)—a feature often missing in survey data. On the negative side, Big Data is often univariate and precludes most forms of quasi-experimental modeling, requiring highly dimensional data.

3. CASE STUDY: HOUSING UNIT SQUARE FOOTAGE DATA

The theoretical framework is useful only insofar as it may be applied in practice. In this section, we apply the TEF to an example for measuring housing unit size using survey data collected via an interviewer, self-reported by a respondent, and scraped from three administrative databases that ideally cover all buildings in the United States. This example was chosen for two reasons. First, it illustrates some of the issues involved in evaluating and contrasting data quality for two or more data sources. In practice, evaluation studies seldom have access to truth or so-called gold standard values, and researchers are tasked with comparing data quality using fallible measurements, as in this study. None of the analyses presented in this section are perfect measures of the extent of error, but combined, they help form a picture of the strengths and weaknesses of the respective data sources and outline one way in which the TEF can be applied.

Second, while square footage itself may not be relevant to most researchers, it is nonetheless demonstrative of similar challenges researchers are encountering with their own data. Given resource constraints, concerns about respondent burden, and desire to reduce measurement error, the Energy Information Administration (EIA) was interested in exploring alternative data collection methods, including the feasibility of replacing or supplementing some survey data with Big Data.

For expository purposes, we limit our consideration to six of the TEF error sources which we believe will provide the greatest contrast between the survey and Big Data: specification, coverage, sampling, nonresponse/missing data, measurement/content, and data processing. This is not to suggest that these are the most important sources of error. Modeling/estimation error and analytic errors were not assessed because they depend on the methods used for combining the RECS data and the Big Data for producing estimates and making inferences. These stages of the inferential process were beyond the scope of the present study, which focused solely on contrasting the quality of the two sources of data.

3.1 Data

3.1.1 2015 Residential Energy Consumption Survey. The Residential Energy Consumption Survey is a periodic in-person, computer-assisted personal interview (CAPI) survey of households that collects energy characteristics, energy usage patterns, and household demographics. The survey has been conducted by the Energy Information Administration since 1978, with the most recent completed iteration in 2015. Unlike other years, it became necessary to switch data collection modes halfway through 2015 data collection due to low response to CAPI. Thus, all incomplete CAPI cases were moved to web and mail modes. For the purposes of this article, we limited analysis to data collected via CAPI because the interviewer assessments of square footage are a key component of the total error evaluation. Thus, the data from web and paper responses were not part of this analysis.

The foundation of the multistage, area probability sampling design relied on the United States Postal Service Computerized Delivery Sequence File (CDS) but was supplemented by field enumeration and frame enhancement in areas with low CDS coverage. A total of 6,522 sampled housing units were attempted using CAPI, yielding 2,417 CAPI-completed interviews (AAPOR Response Rate 2 of 41.8 percent [AAPOR 2016]).

Square footage was collected in two ways. Respondents were first asked to report their square footage. Then, with respondent consent, the interviewer and respondent derived the square footage based upon actual measurements of the housing unit. Our analysis uses two interviewer-derived estimates of square footage: total square footage and main square footage. The main square footage includes standard living space, whereas the total includes the main areas in addition to heated or cooled attached garages, all basements, and finished, heated, or cooled attics.

3.1.2 The Big Data sources. After a review of a variety of data sources, square footage data was obtained from three external databases: Zillow.com, Acxiom, and CoreLogic.[2] The various databases compile data from other sources. Zillow.com and CoreLogic obtain their data from county, municipal, and jurisdictional records. Zillow.com entries may be directly edited by realtors or owners or may be edited by Zillow.com based on realtor listings found on other sites. Acxiom's data sources are less transparent. We only know that they obtain their data from various public records.

After data collection was complete, we attempted to match the addresses of CAPI respondents to each of the Big Data sources. Matches were attempted via Zillow.com's free API tool in the R programming language. Acxiom matching was performed by RTI International. Finally, CoreLogic matched the addresses to their database. In all three databases, an address had to match the exact street address, unit number, city, and state to count as a match. In the case of Zillow.com, we also attempted to match the full CAPI sample (i.e., respondents and nonrespondents); this was not possible for Acxiom or CoreLogic.

3.2 Applying the TEF

3.2.1 Coverage error. For the purposes of RECS, coverage error was defined as the extent to which the enhanced CDS frame did not cover all US housing units or included nonhousing units (e.g., businesses) and duplicate addresses. Unfortunately, no quantitative analysis was possible for this error source because a perfect list of all housing units was not available. However, we believe coverage error is small for three reasons. First, interviewer listing and other frame enhancement procedures were used in sampled segments for which the CDS was suspected to have poor coverage.[3] Given this enhancement protocol,

2. These databases were chosen for their diversity, cost, and a preliminary evaluation of coverage.

3. Poor coverage was defined as any segment for which the count of CDS residential addresses that geocoded into a segment was less than 90 percent of the American Community Survey housing unit count.

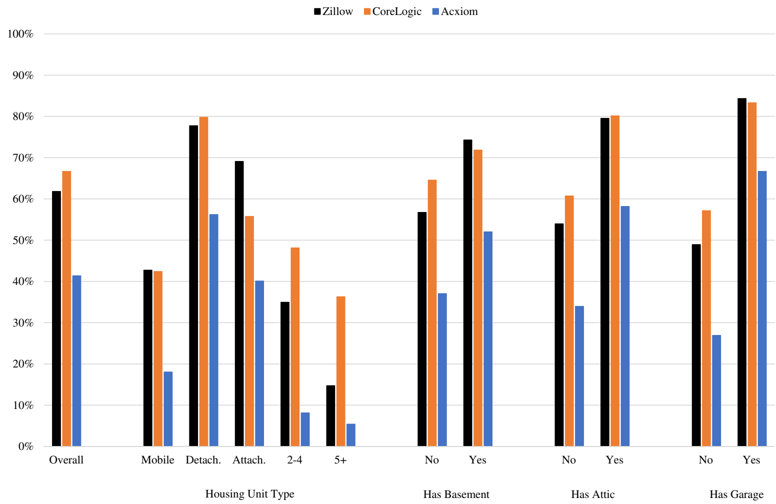

Figure 2. Match Rate by Big Data Source and Key Subgroups.

there was no reason to suspect that any remaining uncovered units would be different from covered units. As a result, the RECS frame should have suffered minimal undercoverage and minimal, if any, coverage bias.

Second, throwback addresses[4] and PO boxes were excluded from the frame prior to sample selection. PO boxes can be opened by anyone, and individuals with a PO box frequently also have another address. Eliminating these types of addresses minimized the risk of duplicates or overcoverage.

Third, interviewers were provided with an extensive list of disposition codes and detailed training on how to identify nonhousing units. Businesses and other nonhousing units were appropriately detected and excluded during data collection to further reduce overcoverage.

Unlike the survey, the third-party records have high levels of coverage error due to undercoverage, introducing coverage bias and (minimal) overcoverage. Undercoverage was calculated as the proportion of responding units for which a match was not found. A match was unavailable for 22.1 percent of responding households, whereas all three Big Data sources had matches for 30.4 percent of cases, and the remaining 47.5 percent of cases matched to one or two data sources. CoreLogic provided the highest match rate (66.7 percent), whereas Acxiom provided the lowest (41.4 percent) (figure 2). While some unmatched cases may be the result of processing error due to data linkage mismatches, the majority is likely due to coverage error. Zillow.com claims to contain data on 110 million addresses, whereas Census reports an additional

4. Throwbacks are addresses associated with a housing unit but delivery is forwarded to another address.

25 million housing units nationwide, suggesting a net coverage rate of approximately 82 percent.

Coverage bias was evaluated in two ways. First, we assessed whether there was a difference in coverage by covariates of housing unit size. All three Big Data sources were more likely to have listings on single family homes (especially detached single family homes) and those with basements, attics, and/or garages—correlates of larger housing units (cf. figure 2).

Second, we compared the average total square footage (as measured by the interviewer) among responding units with a Big Data match to those without any Big Data Match using a two-sample *t* test. While the total square footage may suffer from its own sources of error, the survey error should not be correlated with the Big Data measurement/content error. Observed differences between matched and unmatched cases suggest (although does not prove) the presence of coverage bias among the Big Data. Addresses for which a third-party record was available were measured to be 165 square feet larger on average than units without a match (2,154 versus 1,989, $p < 0.001$).

A thorough assessment of overcoverage would have required a comparison of all Zillow.com, CoreLogic, and Acxiom data to the full (and true) set of housing units. We did not have these data nor the computing power to process them. However, anecdotal evidence suggested some level of duplication existed in each of the Big Data sources. For example, 123 Fernández Street may also be listed as 123 Fernandez Street (no accent mark). The information stored in both records should be the same, creating duplicate entries. Nonhousing units, such as deeded parking spaces, were also observed on the Zillow.com frame as apartments (e.g., 123 Main Street, Unit P1).

3.2.2 Sampling error. The RECS is designed to satisfy specified precision requirements and is therefore adequate for the primary objectives for the survey. However, compared with the extensive data available from the Big Data sources, the effective sample size based on the number of RECS respondents may be considered quite small. The number of respondents for the full RECS sample (including both CAPI and mail/paper components) is 5,686, while the Big Data sources potentially cover at least two thirds of all US housing units.

Our evaluation study was limited to small sample sizes by design because we restricted the Big Data cases to RECS respondents. However in actual practice, it would be feasible and highly desirable to use the entire Big Data file, possibly in conjunction with the survey data, to produce estimates of square footage with adequate precision. Therefore, for future applications of unified Big Data and sample survey data sources for the estimation of square footage, the only sampling error that would apply is that imparted by the sample survey (RECS in the current application). Thus, we believe this level of sampling error should be acceptable.

3.2.3 Specification error. In our example, specification and measurement/content error were difficult to parse. If the respondent estimate, interviewer-derived measurement, and Big Data reports were consistent for a given individual, then we could conclude that there is negligible specification and measurement error. However, if any of these sources diverged, the cause was difficult to deduce. The difference could result from a definitional mismatch—EIA's definition of square footage may be different from the respondent's. Or, it could be the result of measurement/content error—the Big Data source may not have been updated since a home addition was completed. For the purposes of this illustration, any differences among estimates were attributed to measurement/content error and are discussed in that section. Other researchers may have the opportunity to better parse these errors or may attribute the findings differently.

We concluded that the interviewer-derived measurements suffered negligible specification error. Prior to data collection, interviewers were required to complete a home study, in-person training and pass a certification exam. During data collection, they received additional coaching and two newsletters that communicated helpful tips. Moreover, interviewers had a worksheet that provided explicit instructions on how to measure the housing unit and what to measure. These data were then combined at the central office, nearly guaranteeing that the correct areas were included in the estimates.

While specification error was likely low in the interviewer measurements, the opposite is likely true for the respondent estimate. Respondents were asked, "To understand the usage of energy in your home, we need to know about its size and shape. About how many square feet is your home? Your best estimate will do."

No additional help text or instructions on what to include in their estimates was included, leaving them to derive their own definition of square footage. It is unlikely that respondent definitions were consistent with EIA's official definition: include main living space in addition to heated or cooled attached garages, all basements, and finished, heated, or cooled attics. We hypothesize (but cannot confirm) that respondents are more likely to report the taxed, appraised, or realtor-provided estimate. All of these estimates are different but often only include livable area (i.e., not garages and often not basements).

Similar to the respondent data, definitions of square footage were not available for data from Zillow.com or Acxiom, making an assessment of specification error difficult. However, CoreLogic included a variable on the area covered in the estimate. Values included building, living area, adjusted area, or gross area. Only multifamily units could have a building estimate. A building estimate included square footage of the entire building, not just the unit of interest. Explicit definitions for living area, adjusted area, and gross area were not available, but the differences were small and represented differences between official records such as deeds and county tax records. About 11 percent of CoreLogic estimates were only available at the building level, presenting a clear specification error.

Additional specification error was likely but undetectable given the available data. It is likely that property record information more closely aligns with the definition used to construct the main living space measurement (see figures 4A–4C for evidence of this). We suspected that tax and real estate records (the source of most information in the databases) used an alternative definition of square footage. Unlike the EIA definition of square footage, tax records likely do not calculate square footage based on which spaces are heated and cooled.

3.2.4 Nonresponse/missing data error. We assessed nonresponse error of the survey data in three ways. First, unit nonresponse was high for CAPI—36.5 percent of housing units did not complete the survey. Second, item nonresponse on our critical item, housing unit size, was also high for CAPI—26.6 percent for the respondent reports and 19.1 percent for the interviewer measurements.

Third, we assessed nonresponse bias. To assess bias, it is necessary to have a square footage estimate for both RECS respondents and nonrespondents to CAPI. We had Zillow.com square footage values for 1,719 RECS respondents (71.1 percent) and 2,357 RECS nonrespondents (70.0 percent). While the average Zillow.com estimate for any given case may suffer from other sources of bias, there was no reason to suspect that error in the Zillow.com data should be correlated with survey unit nonresponse. The average Zillow.com reported housing unit size for CAPI cases in the full sample (nonrespondents and respondents, combined) was 80 square feet higher than the Zillow.com average among RECS respondents to CAPI, suggesting the presence of sizeable nonresponse bias in the RECS CAPI data (1,719 versus 1,799, $p < 0.01$).[5]

Only item-level missingness was considered for the three Big Data sources because, as previously noted, unit missingness is confounded with undercoverage error in the external data sets. There was no missingness among the Acxiom and CoreLogic databases since only addresses with square footage values are included in these data sources. However, 10.3 percent of Zillow.com records suffered from item missingness—the address was in their database, but housing unit size was not available. Given the small number of RECS CAPI respondents, there was not enough power to detect a reasonable level of bias, so no calculations were made.

3.2.5 Measurement/content error. For the purpose of this set of analyses, we required a gold standard or "preferred" measurement (i.e., the measurements

5. Jackknife repeated replication (JRR) was used to account for the lack of independence between the two samples.

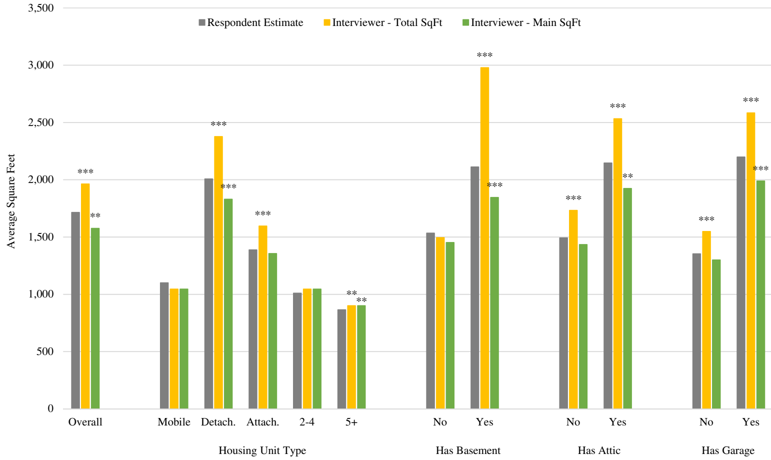

Figure 3. Average Respondent Reported Square Footage Compared with Gold Standard Measurements.

with the smaller absolute bias) of square footage so that measurement biases could be evaluated. The analysis regarded the interviewer-collected measures as the preferred measurements for several reasons. The RECS interviewers were well trained in the collection of square footage data, they measured each housing unit and computed square footage directly, and they applied the RECS definition of square footage consistently across all households. Respondents, on the other hand, do not provide consistent reports of square footage. Some respondents rely on square footage values from property tax forms or a variety of other sources that may not conform to the EIA definition and are, therefore, often inaccurate. Thus, we will interpret the difference between the interviewer value and respondent value as an indicator of bias in the respondent report. In reality, however, this difference is actually measuring the differential bias in the two reports and will only reflect respondent measurement bias to the extent that the interviewer absolute bias is close to zero.

Based on this assumption, we assessed measurement error by calculating (1) the bias and (2) the reliability of the respondent estimate.[6] On average, respondents estimated their housing units to be 1,715 square feet, significantly lower than the interviewer total square footage measurement (1,964, $p < 0.0001$) and significantly higher than the main living space measurement (1,576, $p = 0.002$) (figure 3). This trend held constant across most subgroup comparisons, suggesting significant bias.

6. For each comparison, analysis was limited to the set of cases that had values from the relevant two sources. As a result, the interviewer measurements and the case counts vary slightly across comparisons.

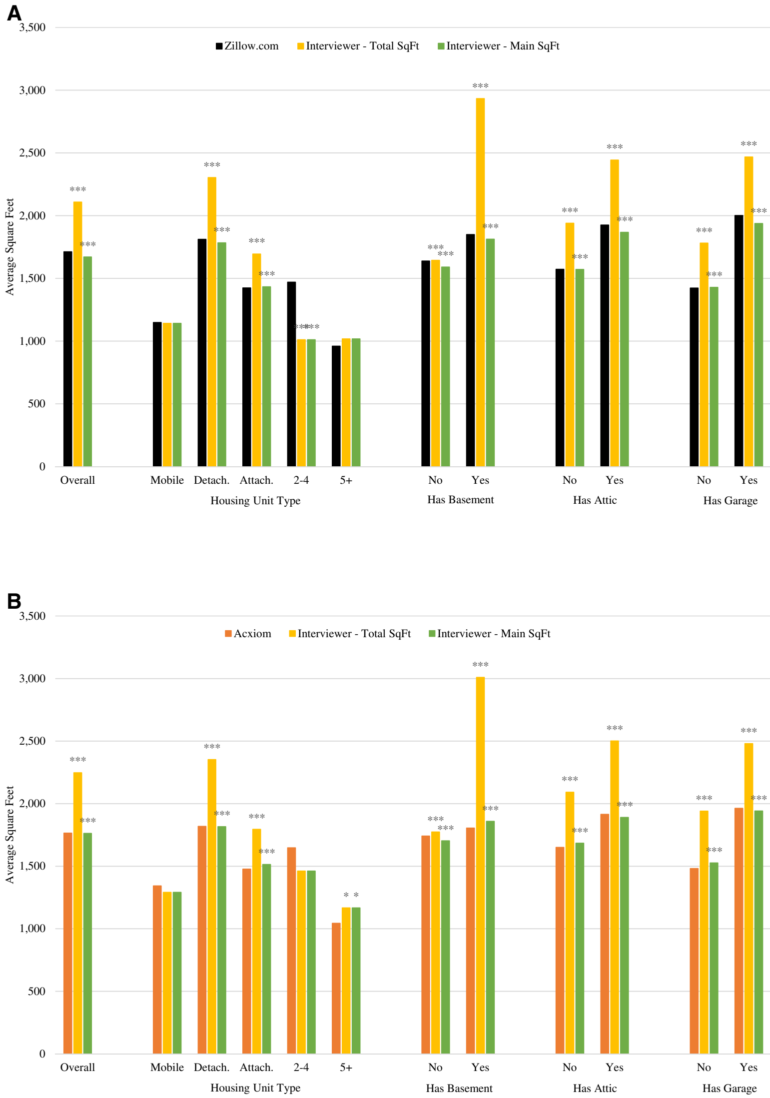

Figure 4. (A) Average Zillow.com reported square footage compared with interviewer measurements. (B) Average Acxiom reported square footage compared with interviewer measurements. (C) Average CoreLogic reported square footage compared with interviewer measurements.

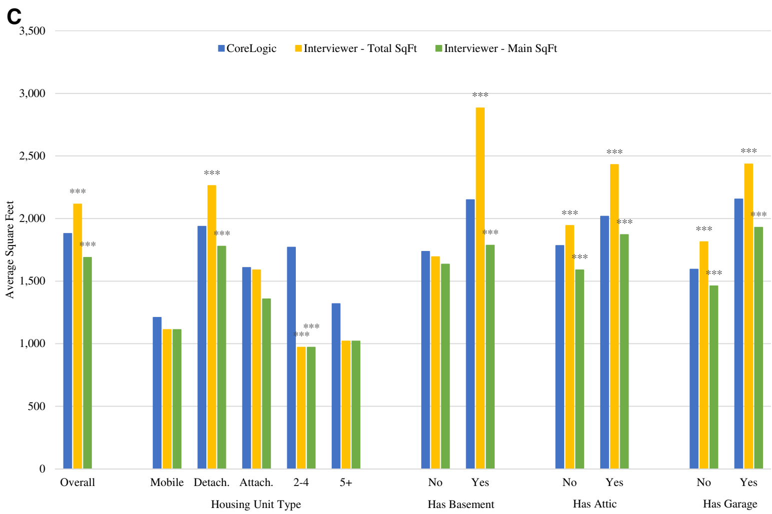

Figure 4. Continued.

A few exceptions existed. For example, we failed to detect bias among some groups such as mobile homes. However, the reliability of the respondent estimate compared with the interviewer total square footage measure was only 49.8 percent. In other words, 50.2 percent (1–0.498) of the variance of the respondent estimate is the result of measurement error and does not represent the variance of the gold standard. This finding was not unique to mobile home dwellers. In the overall estimates, the reliability was 62.4 percent. A scatter plot (not shown) also supported this finding. While a linear trend between respondent and the interviewer measurements existed, it was far from exact.

We applied the same analytic approach to determine the extent of error in the Big Data. Figures 4A–4C display the results. The average unit size for the Zillow.com estimate was 1,713 square feet, significantly lower than the total square footage measurement (2,109 square feet, $p < 0.0001$). Lower Zillow.com estimates persisted across subcategories with few exceptions. The average Zillow.com housing unit size estimates were much more similar to the main living space measurements. Nearly half of the comparisons were no longer significant at the 0.05 level.

Despite Zillow.com's relatively close approximation of the average unit size compared with the main living space, the reliability of the data was 67.2 percent. This is slightly higher than the reliability of the Zillow.com data when compared with the total square footage (65.1 percent) but is still low. The variability of the Zillow.com data suggests large measurement error in addition to the previously identified specification error. Similar findings were identified in the Acxiom and CoreLogic data. Realtors, government surveyors, contractors, and others may incorrectly measure or may guess at the housing unit size. To the extent that these estimates are incorporated into the Big Data sources, they introduce measurement error.

An additional analysis among the Big Data sources further suggested the presence of measurement error. Individual values among the three sources frequently differed for a given case. A paired *t*-test among cases that had a Zillow.com and Acxiom value suggested significant differences ($p < 0.0001$). Additional tests determined that Zillow.com and Acxiom were both significantly different from CoreLogic estimates ($p < 0.0001$ in both comparisons). CoreLogic estimates were generally higher than Zillow.com and Acxiom and were typically closer to the total square footage.

3.2.6 Processing error. Unfortunately, interviews were not recorded, so there was no way to assess the frequency with which interviewers entered a value different from what the respondent reported. However, we expect the number to be low since checks were placed in the CAPI system to confirm implausible and inconsistent responses. The processing of square footage data for RECS followed standard survey data processing procedures (for example, editing of implausibly small or large values) and was expected to contribute minimal error risks.

Two sources of processing error were evaluated for the Big Data: data cleaning and data linkage. For all three Big Data sources, the third-party values were compared with the interviewer measurements, and a set of decision rules were developed to discard values outside of "reasonable" ranges. Reasonable ranges varied by housing unit type. For example, mobile homes with values less than 100 and greater than 3,000 square feet were erased. More data were available for CoreLogic estimates such as the type of space for which the estimate was supposed to cover (e.g., the whole building or just a unit, in the case of apartments) and some area-specific data (e.g., basement square footage). Where discrepant information existed within the CoreLogic variables, data on the overall square footage estimate were edited.

Less than 1 percent of Zillow.com and Acxiom data were erased, whereas 23.5 percent of CoreLogic estimates were altered. This was primarily because CoreLogic frequently provided square footage values for apartment buildings and not individual units within the building. It is impossible to assess the level of processing error (i.e., how many edits were made incorrectly), but analysis (not shown) suggested all edits moved the average square footage values closer to the total square footage. As a result, we concluded that the cleaning procedure improved the accuracy of the estimates, and processing error was low.

Linkage errors occurred when a row for a given address existed in the Big Data source but was not identified and extracted for analysis. This could occur if the address listings were not identical (e.g., a search for "123 Main St" would not identify "123 Main Street"). This error was unlikely to occur when matching the sample frame address to Acxiom or CoreLogic since all three sources used the same standardized United States Postal Service address format. Zillow.com does not standardize its addresses. To determine the rate of mismatch, we drew a subsample of 100 unmatched Zillow.com addresses (11 percent of the total) and manually searched alternative street spellings, abbreviations, and longitude/latitude coordinates. Only two additional matches were found, suggesting a mismatch rate of 2 percent.

3.3 Error risk profile. In this section, we use the results of the TEF evaluation and our knowledge of the error risks associated with both survey data and the Big Data sources considered in this article to compare the error risks associated with square footage data derived from a full-scale RECS CAPI survey and a hypothetical square footage data set derived from Big Data sources. This comparison is shown in table 2.

The risk levels in table 2 pertain specifically to square footage data as collected by RECS CAPI data (for the survey column) and use external data sources such as Zillow.com for the Big Data component. Where evaluation data were not available to provide data-driven assessments, we substituted speculative assessments until such data become available. These speculative risk assessments are identified in the table in regular font, and the data-driven assessments are denoted by bold font.

As mentioned previously, the 2015 RECS ultimately included a mixed mode design, moving CAPI nonrespondents to web and mail modes. Because data collected in these self-administered modes did not include an interviewer measurement of square footage, they were excluded from this analysis. However, inclusion of these cases in this analysis would have reduced unit nonresponse and may have altered the extent of nonresponse bias identified in the RECS data. Similarly, we limited analysis of the Big Data sources to RECS CAPI respondents, but a true use case would likely have included the full RECS sample or a larger subsample of it. While an expanded sample of the Big Data sources is less likely to have altered the risk assessment in table 2, it may have introduced the potential for additional analyses such as small area estimation. The assessment found in table 2 should not be considered a final evaluation of either data set but an example of the applicability of the TEF.

4. DISCUSSION

Big data offers the potential for a variety of research to be conducted faster and at a lower cost than surveys allow. But these data need to be used responsibly, and their limitations need to be understood and accounted for. To date, many researchers have evaluated the validity and/or reliability of the final estimate (e.g., Google Flu Trends [Ginsberg et al. 2008; Cook et al. 2011; Butler 2013]). This is an excellent starting point, but insight into the components of

Table 2. Summary of Analysis and Findings of the TEF Evaluation for RECS Square Footage

TEF error component	Survey data			Big Data		
	Reason for error	Analysis	Assessed level of error	Reason for error	Analysis	Assessed levelof error
Coverage	Undercoverage	Qualitative evaluation of frame construction methods Qualitative inference given undercoverage conclusion	Minimal	Undercoverage	**Match rate** **Compared match rate by covariates of size and *t* test between matched and unmatched**	High
	Overcoverage	Qualitative evaluation of interviewer methods	Minimal	Overcoverage	Anecdotal evidence	Minimal
	Duplication	Qualitative evaluation of interviewer methods	Minimal	Duplication	Anecdotal evidence	Minimal
Sampling	Multi-stage unequal probability sampling	Standard error calculations	High (relative to Big Data)	Complete census of data source is possible	Census data are not subject to sampling error	N/A
Specification	Concepts difficult to operationalize	**Overlaps w/measurement error; *t* tests between respondent estimate and interviewer measurements**	High	Concepts inconsistent with available data	**Overlaps w/ measurement error; *t* tests between Big Data and interviewer measurements**	High

Continued

Table 2. *Continued*

TEF error component	Survey data			Big Data		
	Reason for error	Analysis	Assessed level of error	Reason for error	Analysis	Assessed levelof error
Nonresponse/missing	Unit nonresponse	**Response rate t test between full sample and respondents**	Moderate	Unit nonresponse	**Match error**	Minimal
	Item nonresponse	**Item-level nonresponse**	High	Item nonresponse	**Item-level nonresponse**	Zillow.com: Moderate; Others: N/A
Measurement/content	Unintentional error Deliberate error	**Unintentional and deliberate indistinguishable; t tests and reliability of respondent estimate versus interviewer measurements**	High	Unintentional error Deliberate error	**Unintentional and deliberate indistinguishable; t tests and reliability of Big Data versus interviewer measurements**	High

Processing	Editing error	Qualitative evaluation of data processing procedures	Minimal	Editing error	**Proportion of data edited** **Compared the differences between edited values and interviewer measurements to the differences between raw values and interviewer measurements using paired *t* tests**	Low
	Keying error	Qualitative evaluation of data collection methods	Minimal	Linkage error	**Manually searched alternative addresses for a subsample of unmatched addresses**	Minimal

the error are critical. Only by assessing the components can researchers understand how changes to what data are made available to researchers, the definition of a variable, or how a researcher downloads the data may impact the stability or bias of an estimate. Similarly, by breaking down error into its components, we can identify where the smallest investment may make the largest impact on how survey and Big Data may be used in tandem to produce a higher quality estimate.

The goal of this article is to provide a framework to serve as a guide for researchers to enumerate the various types of error possible in Big Data. We propose researchers consider error from eight components: (1) coverage/domain, (2) sampling, (3) specification, (4) nonresponse/missing data, (5) measurement/content, (6) processing, (7) modeling/estimation, and (8) analytic. The Total Error Framework mimics the Total Survey Error Framework in that we seek to provide scientists with a tool that can be used to consider different components of the research process that may introduce error in the final estimates or models. Where possible, we have adopted the TSE terminology for the TEF to make it easier to communicate this new framework to social scientists and survey methodologists. However, there are two primary differences between the TSE and TEF. First, to maximize accessibility of the TEF, we have provided examples across the spectrum of Big Data sources to ensure researchers can conceptualize how this framework may apply to their specific (and often unique) data source. Second, as demonstrated in figure 1 and unlike the TSE, the TEF is not linear. The extraction, transformation, and load processes often happen in tandem or through an iterative process. Whereas researchers may consider TSE error components one at a time in a stepwise progression, TEF components may need to be considered all at once, backward, or not at all. Ultimately, the value of the TEF is its flexibility. While we have demonstrated how the TEF can be applied to square footage data, its applicability is generalizable to other data sets whether they include data from wearable devices (e.g., FitBit), video (e.g., traffic patterns), or social media (e.g., Tweets).

However, the application of the TEF is not easy. As evidenced by the square footage illustration, the information required to conduct the ideal investigation is rarely, if ever, available. In some cases, this is because the truth may be unknowable, the information is proprietary, or the effort required would be cost prohibitive. Instead, we must use what is available and attack the problem from several different angles, relying on the resulting big picture as opposed to the individual tests. Under this approach, we found that neither the respondent-provided survey responses nor the Big Data provided reliable or unbiased estimates of housing unit size, but the sources of error varied. This information may be used in combination with the researcher's priorities to determine which data source to use. For example, if our goal is to estimate the average building size in the United States, we may opt to use the survey data since it has significantly less coverage bias. However, if we are using square footage as an input in a model to estimate energy consumption, we might choose the administrative data. In the latter scenario, the bias of the building size may be unimportant as long as the correlation between building size and energy consumption is consistent (something untested in this article).

One weakness of this article (and both the TSE and TEF) is that it only considers data quality. Several other factors should be considered when deciding whether to supplement or replace survey data with Big Data. These include but are not limited to cost, timeliness, and risk. While we expect readers to have a sound understanding of the expenses incurred with surveys, Big Data can also be expensive, reducing the cost savings expected from Big Data. Gaining access to a given data source may require data use agreements that must first be reviewed by lawyers and contracts officers, increasing costs and the time required to acquire the data. Similarly, Big Data can often take up terabytes of space and significant memory to analyze. Special servers may need to be purchased and maintained. These data are also created outside of a secure environment and may need to be scrubbed for personal identifying information and reviewed for viruses prior to being stored on an organization's servers.

Big Data is typically created by a third party (e.g., Twitter or Zillow.com), which introduces significant risk when used. If the third party is defunct, changes what data are collected, how frequently they are collected, or how accessible they are to third parties, then the ability to use them may be jeopardized. Third parties often change the way in which their data are collected, which may disrupt time series created from these data or may change the validity of the estimates produced. For example in 2017, Twitter changed its character limit from 140 to 280 per tweet. Researchers often use sentiment analysis to extract social attitudes from tweets, but these models are less effective when used on very short excerpts. The longer character limit may affect the noise in sentiment models, the strength of sentiment, or even the presence of detectable sentiment, making it appear as if the twitterverse is becoming more opinionated. While Twitter's change in character limits is a public example of change to Big Data source, several other changes may be proprietary or so far removed from the researcher that they are never hypothesized to have an effect on the outcome of interest. It will be critical for researchers to understand how transparent the third party is willing or able to be with the collection and maintenance of their data and to consider how changes may affect results.

A second weakness lies in the examples used in this article. With minor exceptions, we have used examples that assume analysis using rectangular data files (i.e., rows and columns). As analytic methods improve and—more importantly—become more widely utilized, we will need to consider whether the TEF can incorporate these nontraditional data formats (which we believe it can) or whether it will need to be expanded upon.

A final consideration for researchers will be the evaluation of error when combining survey and Big Data. Understanding the error sources independently for Big Data and survey data is the first step. However, if these data are combined, it will also be important to understand how different harmonization methods affect the resulting estimates and models. While we do not address data combination here, the TEF would still be applicable, as the method for merging multiple data sources could be considered part of the processing error component.

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